Sujit Silas Armstrong Suthahar

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RESEARCH EXPERIENCE

UCLA Dermatology and Immunology **Los Angeles, CA**

Graduate Student Researcher III *Apr 2025 – present*

- Integrated public and in-house scRNA-seq datasets with scvi-tools into a mouse skin wound-healing atlas, creating the reference used for downstream macrophage and epithelial state analysis
- Compared burn and sham macrophage states in scRNA-seq data, identifying a rare dysfunctional macrophage population
- Developed a VAE-based causal inference framework with adversarial disentanglement and optimal transport to estimate individual treatment effects from ~57K-cell scRNA-seq data, enabling in-silico gene perturbation and counterfactual prediction across cell types and timepoints
- Built a doubly robust estimator for scRNA-seq ambient RNA decontamination with identification assumptions, empirical Bayes local soup estimation, and per-cell stochastic posterior decomposition, benchmarked against CellBender and SoupX

UCLA Institute for Society and Genetics **Los Angeles, CA**

Graduate Student Researcher I *Sep 2023 - Mar 2025*

- Led multi-omics analysis (RRBS, WGBS, ATAC-seq, RNA-seq) to elucidate epigenetic effects of environmental toxins
- Applied PARAFAC AO-ADMM based CMF to uncover shared latent factors across biological modalities [preprint]
- Developed a genome-wide differentially methylated region annotation pipeline integrating chromatin state and annotations
- Designed Nextflow pipelines for reproducible bioinformatics workflows on UCLA's Hoffman2 HPC cluster

UCLA Biomedical AI Research Lab **Los Angeles, CA**

Researcher *Sep 2023 - Jun 2024*

- Designed a multiple instance learning classifier for thyroid nodules, achieving an 11% sensitivity improvement [published]
- Developed U-Net based segmentation model with enhanced skip connections for biopsy needle segmentation
- Applied transfer learning with pre-trained VGG-16 backbone to extract features from ultrasound and CT images, enabling efficient model training on limited medical imaging data

WORK EXPERIENCE

Aithrel

Independent AI Engineering Projects *Apr 2025 - Oct 2025*

- Developed an agentic RAG system with LangGraph for document retrieval and knowledge extraction tasks
- Built an in-house MCP server giving agents structured, permissioned access to DynamoDB
- Created a model-serving endpoint on Databricks with OAuth to run a multimodal document processing pipeline on serverless compute, achieving low-latency inference (2-3s per document) across 1,000+ engineering documents
- Hosted open-source multimodal embedding models on Databricks to serve image retrieval for engineering document sets

GRAIL **Menlo Park, CA**

Computational Biology & Machine Learning Research Intern *Jun 2024 - Sep 2024*

- Optimized feature selection for argmax classifier improving cfDNA probe specificity for 150+ probes on the Galleri targeted assay panel
- Designed a genome annotation toolkit for DMRs, deployed internally and re-annotated 400+ datasets
- Refactored analytical functions and merged pull requests to internal codebase

La Jolla Institute for Immunology **La Jolla, CA**

Bioinformatics Specialist *Jun 2021 - Jul 2023*

- Led RNA-seq, scRNA-seq, CITE-seq, and TCR-seq analyses using DESeq2, edgeR, Scanpy, Seurat, and ad hoc pipelines
- Designed statistical models for analyzing 348,000+ TCR sequences, identifying 672 APOB-reactive T-cell clones and 532 conserved CDR3 motifs to track disease-specific clonotypes across donors
- Developed a computational toolkit for optimizing titration of 124 antibodies in CITE-seq; published in Scientific Reports
- Automated sequencing data alignment and analysis pipelines reducing processing time by 30%

TECHNICAL SKILLS

Programming: Python, R, TypeScript, Bash/Shell, SQL, C/C++, LaTeX
ML/DL: PyTorch, Scikit-learn, JAX, TensorFlow, Keras, LangChain, LangGraph, Transformers, Hugging Face
Cloud & Infra: AWS (EC2, Lambda, S3, DynamoDB), Docker, Git, CI/CD, MLOps, Databricks
Data Analytics: Pandas, NumPy, Apache Spark, ETL Pipelines, Matplotlib, Seaborn, Plotly
Bioinformatics: DESeq2, Seurat, Scanpy, Nextflow, Cell Ranger, Bioconductor, WGCNA, MiXCR, edgeR

PUBLICATIONS

Peer-Reviewed Journal Articles (First Author / Co-First)

- **Suthahar SSA**, Roy P, Makings J and Ley K (2024) Identification of apolipoprotein B–reactive CDR3 motifs allows tracking of atherosclerosis-related memory CD4+T cells in multiple donors. Front. Immunol. 15:1302031. doi:10.3389/fimmu.2024.1302031.
- **Armstrong, SS,** Chen, DG, Kumar, S, et al. (2025). CITE-Seq Analysis Reveals a Differential Natural Killer Cell SPON2 Expression in Cardiovascular Disease Patients. International Journal of Molecular Sciences, 26(3), 1369.
- **Armstrong Suthahar, SS**, Nettersheim, FS, Alimadadi, A, et al. (2024). Olfr2-positive macrophages originate from monocytes proliferate in situ and present a pro-inflammatory foamy-like phenotype. Cardiovascular Research, 120(13), 1577-1589.

Preprints (First Author)

- **Suthahar, SSA**, Allard, P. (2025). Integrative Analysis of Epigenetic, Transcriptomic, and Metabolomic Responses to Arsenic Exposure Using Coupled Matrix Factorization. arXiv:2510.19294. [Submitted]
- **Armstrong, SS**, Ruan, D. (2025). Generation of synthetic scRNA-seq-like transcriptomes using a GAN from RNA-seq data. bioRxiv, 2025.10.09.681449.

Selected Co-Authored Publications

- Tada, Y, **Armstrong, SS,** et al. (2025). Proinflammatory and cytotoxic CD38+HLA-DR+ effector memory CD8+ T cells are peripherally expanded in human cardiac allograft vasculopathy. American Journal of Transplantation.
- Roy, P, Bellapu, A, **Suthahar, SSA,** et al. (2025). Loss of effector Treg signature in APOB-reactive CD4+ T cells in CAD. Nature Cardiovascular Research, 4(7), 841-856.
- Nettersheim, FS, Brunel, S, Sinkovits, RS, **Armstrong, SS**, et al. (2025). PD-1 and CD73 on naive CD4+ T cells synergistically limit responses to self. Nature Immunology, 26(1), 105-115.
- Athreya, S, Melehy, A, **Suthahar, SSA**, et al. (2025). Combining Ultrasound Imaging and Molecular Testing in a Multimodal DL Model for Thyroid Nodules. Thyroid, 35(5), 590-594.
- Oliaeimotlagh, M, Kumar, S, Taraskin, A, **Suthahar, SSA**, et al. (2025). Automated denoising of CITE-seq data with ThresholdR. Cell Reports Methods, 5(7).
- Freuchet, A, Roy, P, **Armstrong, SS**, et al. (2023). Identification of human exTreg cells as CD16+ CD56+ cytotoxic CD4+ T cells. Nature Immunology, 24(10), 1748-1761.
- Nettersheim, FS, **Armstrong, SS**, Durant, C, et al. (2022). Titration of 124 antibodies using CITE-Seq on human PBMCs. Scientific Reports, 12(1), 20817.
- Ulaganathan, G, Jiang, H, Canio, N, Oke, A, **Armstrong, SS**, et al. (2024). Screening and characterization of 133 physiologically-relevant environmental chemicals for reproductive toxicity. Reproductive Toxicology, 126, 108602.
- Iqneibi, S, Saigusa, R, Khan, A, Oliaeimotlagh, M, **Armstrong Suthahar, SS**, et al. (2023). Single cell transcriptomics reveals recent CD8 T cell receptor signaling in patients with coronary artery disease. Frontiers in Immunology, 14, 1239148.

* Please check my Google Scholar page for information on additional publications

Poster Presentations & Published Abstracts (Most Recent)

- **Armstrong Suthahar, SS,** Allard, P. "Integrative Multi-Omics Analysis of Arsenic Toxicity Using CMF." Society of Toxicology (SOT) March 2025, Orlando, FL.
- Shreeram Athreya, Andrew Melehy, **Armstrong Suthahar, SS**, "Multi-Modal Deep Learning for Thyroid Nodule Risk Stratification." UCLA JCCC Symposium, Mar 2024.
- **Armstrong Suthahar, SS**. "Generation of synthetic scRNA-seq-like transcriptomes using a GAN from RNA-seq data." UCLA Bioengineering Conference, Feb 2024.

SELECT PROJECTS

- State Model Experimentation 2026
  Evaluated Arc Institute's State Embedding (SE) and State Transition (ST) models by conditioning them on time-course data to capture temporal patterns in skin wound scRNA-seq experiments
- Bio-MCP Server for Single-Cell Omics 2026
  Built an MCP server exposing scRNA-seq analysis as typed, composable tools with handle-based I/O and 4 KB response caps, per-project SQLite memory and provenance, and notebook export so results reproduce. Designed for local Qwen-class (30-35B) models as the tool-caller
- Metastatic Tissue Classification with Grad-CAM 2025
  Benchmarked fine-tuned EfficientNet and InceptionV3 backbones on PatchCamelyon histopathology tiles over 20 epochs, reaching 89.1% validation accuracy, and applied Grad-CAM saliency maps to inspect which tissue regions drove each prediction
- Coupled Matrix Factorization for Multi-Omics Integration 2025
  Developed CMF framework to jointly decompose multi-omics data (methylation, transcriptomic, metabolomic), identifying shared latent factors; preprint on arXiv
- Finetuning Mistral-7B-Instruct 2025

Finetuned Mistral-7B-Instruct-v0.3 with LoRA using the official mistral-finetune framework to build a domain-adapted model for technical question answering

- tcRdiff: R Package for Differential TCR Motif Analysis 2024
  R package extending GLIPH2 output to identify clonally expanded CDR3 motifs using exact Poisson tests on summed template frequencies, with volcano plots for enriched motifs; installable via devtools

ACADEMIC SERVICE AND LEADERSHIP

UCLA Samueli School of Engineering **Los Angeles, CA**
Graduate Student Representative, Faculty Executive Committee (FEC) *Jan 2024 - Dec 2024*

- Led school-wide survey of 200+ students; identified 70% of master's students require a course in research methodology, informing curriculum policy changes
- Ran computational workshops bridging wet-lab and dry-lab divide and provided academic consulting to international students; recognized with Engineering Achievement Award (2025)

Stealth Startup (UCLA Licensed Technology) **Los Angeles, CA**
Regulatory Consultant *Sep 2024 - present*

- Conducted FDA 510(k) regulatory strategy and competitive landscape analysis for an electrical stimulation wound care device, identifying predicate devices and applicable product codes to support pre-submission discussions with the FDA
- Analyzed reimbursement and market access pathways, mapping CPT/HCPCS billing pathways and payer coverage requirements for the device
- Co-authored an NIH Blueprint translational grant application to secure funding for advancing the technology into clinical validation and support the company's first pivotal randomized controlled trial
- Supported early-stage fundraising efforts that secured $100K in seed capital to initiate manufacturing scale-up and pilot clinical studies, while helping establish strategic partnerships through signed LOIs with UCLA and USC
- Helped structure corporate governance and negotiate a patent licensing agreement with UCLA ahead of seed-stage financing

RELEVANT COURSEWORK

BIOSTAT 245 Causal Inference, ECE C247 - Neural Networks and Deep Learning, ECE C239AS - Signals and Systems: Advanced Neural Networks and Deep Learning, BIOMATH 201 - Deterministic Models in Biology, PBMED 209 - Signal and Image Processing, BIOE 275 - Machine Learning and Data Driven Modeling, BIOE 224B - Advanced Imaging Informatics, GEN 242 - Data Analysis in Genome Biology, ECE C243A – Brain Computer Interfaces, BIOL 119 - Introduction to Genomics and Bioinformatics, BIOL 115 - Human Genetics, AAI Courses in Immunology 2025

TEACHING

LS30A: Mathematics for Life Scientists **Los Angeles, CA**
Teaching Assistant *Sept 2026 – Present*

BIOENGR 175/275: Machine Learning & Data-driven Modeling in Bioengineering **Los Angeles, CA**
Lead Teaching Assistant *Mar 2026 – Jun 2026*

AWARDS AND HONORS

- UCLA Cross Campus Innovation Challenge – 2nd Place ($10000) Apr 2026
- UCLA Lowell Milken Institute: Milken-Sandler Prize – 2nd Place ($12500) Apr 2026
- Engineering Achievement Award in Student Welfare, UCLA Jun 2025
- UCR Highlander Union Excellence Award, UC Riverside Mar 2019

EDUCATION

University of California-Los Angeles **Los Angeles, CA**
Ph.D. in Bioengineering *Apr 2025 - present*
GPA: 4.00/4.00

University of California-Los Angeles **Los Angeles, CA**
M.S. in Bioengineering Sep 2023 - *Mar 2025*
GPA: 3.98/4.00

University of California-Riverside **Riverside, CA**
Bachelor of Science, Biology (Honors) *Sep 2017 - Mar 2021*
GPA: 3.60/4.00